Core Java Study Plan Day 11

Programming Questions

1. Write a Java program that accepts two integers from the user and performs division. Handle InputMismatchException, ArithmeticException, and other unexpected exceptions using separate catch blocks. Use a finally block to print "Execution Completed". Demonstrate the execution order when invalid input or division by zero occurs.

2. Write a Java program using nested try-catch blocks to access an array element, divide two numbers, and convert a string into an integer. Handle exceptions at appropriate levels and include finally blocks in both inner and outer try-catch structures. Demonstrate which blocks execute for different exceptions.

3. Create a method withdraw(double balance, double amount) that throws an IllegalArgumentException if the withdrawal amount is zero or negative and an ArithmeticException if the amount exceeds the balance. Use throws in the method declaration, invoke it from main(), and handle each exception separately.

4. Create a custom checked exception named UnderAgeException by extending Exception. Develop a voting application that accepts a user's age and throws UnderAgeException if the age is below 18. Handle invalid numeric input separately and display appropriate messages for eligible and ineligible voters.

5. Create three methods: methodC(), methodB(), and methodA(). Make methodC() convert a string into an integer and declare NumberFormatException using throws. Call methodC() from methodB() and methodB() from methodA(). Handle the exception only in main() and demonstrate exception propagation through the call stack.

6. Write a Java program that reads a text file using BufferedReader inside a try-with-resources statement. Handle IOException and demonstrate that the resource closes automatically when reading finishes or an exception occurs. Include a finally block and display messages to verify the execution order.

7. Create a custom unchecked exception named InsufficientBalanceException by extending RuntimeException. Implement a BankAccount class with deposit() and withdraw() methods. Reject negative deposits, non-positive withdrawals, and withdrawals exceeding the available balance. Handle exceptions in main() and ensure the balance remains unchanged after failed transactions.

8. Write a Java program that can produce ArithmeticException, NullPointerException, and ArrayIndexOutOfBoundsException. Use multiple catch blocks, including a general Exception catch block. Demonstrate why specific exception handlers must appear before broader handlers and explain what happens if the order is reversed.

9. Accept a string containing an integer and use the converted value to determine the size of an integer array. Handle NumberFormatException for invalid numeric strings and NegativeArraySizeException for negative values. Prevent invalid array creation and ensure the program handles unexpected input without terminating abruptly.

10. Develop a student result processing application that accepts marks for multiple subjects. Validate marks, convert string inputs into integers, calculate the average, and reject invalid marks using a custom checked exception. Handle InputMismatchException, NumberFormatException, and ArithmeticException wherever applicable. Use throws for method-level exception declaration, multiple catch blocks for error handling, and finally to display "Result Processing Finished". Ensure invalid data never produces an incorrect result.

Puzzles

1. A farmer has 17 sheep. All but 9 die. How many sheep are left?

2. A clock strikes 6 times in 5 seconds. How long will it take to strike 12 times, assuming equal intervals between strikes?

3. You have a 6-litre jug and a 4-litre jug. How can you measure exactly 2 litres?

4. A man has two coins whose total value is ₹30. One of the coins is not a ₹10 coin. What are the two coins?

5. A room contains a candle, a fireplace, and a gas lamp. You have one match. What do you light first?

Aptitude Questions

1. In how many ways can 4 men and 3 women sit in a row if all the women must sit together?

A) 360 B) 720 C) 1440 D) 2160

2. How many diagonals does a polygon with 12 sides have?

A) 54 B) 60 C) 66 D) 72

3. A card is drawn from a standard deck. What is the probability that it is a face card or an ace?

A) 4/13 B) 5/13 C) 7/26 D) 1/2

4. Two dice are thrown. What is the probability that the product is even?

A) 1/2 B) 5/9 C) 3/4 D) 2/3

5. A bag contains 4 white and 6 black balls. Two are drawn without replacement. What is the probability that one is white and one is black?

A) 4/15 B) 8/15 C) 2/5 D) 3/5

6. A 30-L mixture has milk and water in ratio 4:1. How much mixture should be replaced by water so that the ratio becomes 3:2?

A) 6 L B) 7.5 L C) 8 L D) 10 L

7. Two varieties of coffee cost ₹320/kg and ₹440/kg. In what ratio should they be mixed to get ₹350/kg?

A) 3:1 B) 7:1 C) 9:2 D) 11:3

8. A solution contains 25% salt. If 8 L water is added to 32 L of the solution, what is the new salt percentage?

A) 15% B) 18% C) 20% D) 22%

9. A is the sister of B. B is the son of C. D is the husband of C. E is the daughter of D. How is E related to A?

A) Sister B) Mother C) Aunt D) Cousin

10. Pointing to a boy, Meena said: 'He is the son of my mother's only daughter.' Meena has no sisters. How is the boy related to Meena?

A) Brother B) Son C) Nephew D) Cousin